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Dose dependent effects of transcranial photobiomodulation on blood-oxygenation-level-dependent power in major depressive disorder

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Abstract

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Dan V. Iosifescu: Conceptualization, Data curation, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Writing – original draft, Writing – review & editing. **Katherine A. Collins:** Investigation, Methodology, Project administration, Supervision, Writing – review & editing. **Umit Tural:** Data curation, Formal analysis, Methodology, Visualization, Writing – original draft. **Jacek P. Dmochowski:** Conceptualization, Formal analysis, Methodology, Writing – review & editing. **Naomi Gaggi:** Formal analysis, Methodology, Software, Visualization, Writing – review & editing. **Zamfira Parincu:** Investigation, Project administration, Writing – review & editing. **Anna Peterson:** Investigation, Project administration, Writing – review & editing. **Aura M. Hurtado-Puerto:** Investigation, Project administration, Writing – review & editing. **Maia B. Gersten:** Investigation, Project administration, Writing – review & editing. **Julie A. Clancy:** Investigation, Writing – review & editing. **Kayla M. McEachern:** Investigation, Project administration, Writing – review & editing. **Tarek Sobeih:** Data curation, Resources, Software, Writing – review & editing. **Luis de Taboada:** Conceptualization, Methodology, Resources, Software, Writing – review & editing. **Thaddeus Tarpey:** Formal analysis, Methodology, Validation, Writing – review & editing. **Paolo Cassano:** Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Writing – review & editing.

Ethics declaration

Written informed consent to take part in the study and to publish the article has been obtained from all participants or their legal representatives. The privacy rights of participants have been observed.

This study was performed in compliance with relevant laws, regulatory frameworks and guidelines where the research took place. This study was approved by the NYU Grossman School of Medicine IRB. (Approval No. i20-00217)

The results of this clinical trial and any associated work have been posted in a registry. This clinical trial was registered with number NCT04366258 (clinicaltrials.gov).

Declaration AI

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process.

No generative AI or AI-assisted technologies were used during the preparation of this work.

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Background: Transcranial photobiomodulation (t-PBM) with near-infrared light stimulates mitochondria and may have antidepressant effects. We evaluated dose-dependent effects of t-PBM on the hemodynamic blood-oxygenation-level-dependent (BOLD) power in major depressive disorder (MDD).

Methods: 31 MDD subjects underwent randomly assigned t-PBM sessions, 1/week, in the MRI scanner, with 1) Sham; 2) High dose: pulse wave, irradiance ~300 mW/cm^2; 3) Medium dose: continuous wave, ~300 mW/cm^2; and 4) Low dose: continuous wave, ~50 mW/cm^2 t-PBM (808 nm) was delivered to the prefrontal cortex, bilaterally. fMRI was recorded at 3T before, during, and after t-PBM. We used mixed-effects linear regression to evaluate changes in BOLD power during stimulation, compared to sham. The analysis was repeated for the middle frontal gyrus (MFG), the prefrontal areas irradiated by t-PBM, and the entire brain cortex.

Results: We found similar results in the MFG, the prefrontal cortex directly irradiated, and the entire brain cortex: medium dose t-PBM was associated with a statistically significant increase in BOLD power, whereas low dose t-PBM was associated with a significant decrease in BOLD. There were no significant changes in BOLD power with the high (pulsed) t-PBM dose or with sham. Single administrations of any t-PBM dose did not result in significant changes in depression severity (versus sham). All 3 t-PBM doses were well tolerated.

Conclusion: The acute effect of t-PBM on the hemodynamic BOLD power is robust, dose-dependent, bidirectional, and extends beyond the areas directly illuminated. These findings may provide a reference for future dose selection and clinical efficacy studies.

Keywords

Major depressive disorder (MDD); Transcranial photobiomodulation; Near-infrared radiation; Functional MRI (fMRI); Blood-oxygenation-level dependent (BOLD); Low level light therapy

1. Introduction

Major Depressive Disorder (MDD) is prevalent and associated with significant disability [1,2]. Antidepressant medications are only partially effective and can have burdensome side effects [3,4].

Transcranial photobiomodulation (t-PBM) with near-infrared radiation (NIR) is a safe, non-invasive neuromodulation strategy based on brain exposure to light at specific wavelengths. Most t-PBM light is attenuated through the bone, but a significant fraction (2-4%, dependent on anatomy, wavelength, hair, skin pigmentation) reaches the brain cortex [5,6]. In brain cells, NIR light is absorbed by the mitochondrial enzyme cytochrome *c* oxidase (CCO), stimulating the mitochondrial respiratory chain and leading to increased ATP production [7-9]. In addition, NIR can improve mitochondrial activity by promoting the dissociation of nitric oxide (NO) from the CCO, releasing the binding site for oxygen, and restoring oxidative phosphorylation [10]. The released NO may also act as a local vasodilator [11], explaining the robust effects of t-PBM on increasing cerebral blood flow [12-14].

t-PBM has shown promising early results for the treatment of neuropsychiatric disorders [15], including MDD [16,17]. However, other MDD studies [18] failed to detect an effect,

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possibly explained by the different t-PBM parameters and doses used. There is no consensus on the optimal irradiance for central nervous system (CNS) disorders, a necessary step before large-scale studies demonstrating efficacy. Based on our preliminary data [12], we aimed to evaluate the dose-dependent effects of t-PBM on the hemodynamic fMRI blood oxygenation level dependent (BOLD) power, in MDD subjects. We hypothesized that the effects would be most robust in the middle frontal gyrus (MFG, pre-specified region), which receives the highest intensity of laser light during the procedure. We also planned analyses of dose-dependent BOLD power in the prefrontal cortical areas illuminated by the laser, and in the entire brain cortex. All study methods were pre-specified [19].

2. Methods

2.1. Participants and clinical assessments

This is the first (R61) of a two-phase (R61/R33) experimental therapeutics study, conducted between August 2020 and June 2022 at New York University (NYU), the Nathan Kline Institute (NKI), and the Massachusetts General Hospital (MGH).

After providing signed informed consent, subjects underwent psychiatric screening (MINI, IDS-C), tests for cognition (MoCA), and skin color (NIS-SCS). Participants of all genders, 18-65 years old, were enrolled if they met DSM-5 criteria for MDD (assessed by MINI) and were acutely depressed (IDS-C ≥ 23 at screen and ≥ 20 at randomization). Medications or psychotherapy approved for MDD had to be stable for at least 8 weeks.

Exclusion criteria included a) serious suicide risk, b) history of any or psychotic or bipolar disorder; c) alcohol or substance use disorder, PTSD, OCD or eating disorders, d) history of dementia, traumatic brain injury, or neurological disorders affecting the brain; e) significant cognitive impairment (MOCA <22); f) history of antisocial personality disorder; g) history of significant treatment non-adherence; h) pregnancy or lactation; i) currently undergoing device-based treatment for depression or taking medications for depression other than SSRIs or SNRIs; j) treatment resistance (no response to >2 adequate antidepressant treatments in the current MDD episode); k) history of ECT (12 months), or VNS (lifetime); lifetime treatment resistance to any FDA-approved device treatment for MDD; l) serious, unstable medical illnesses; m) significant abnormal laboratory parameters including urine toxicology; n) uncontrolled hypothyroidism; o) past intolerance or hypersensitivity to t-PBM; p) significant skin conditions on the subject's scalp in the area of the procedure sites; r) any use of light-activated drugs within 14 days prior to study enrollment; s) implants in the head whose functioning might be affected by t-PBM; t) failure to meet MRI safety requirements.

At all follow-up visits (with t-PBM during fMRI), clinical and safety assessments included:

Montgomery-Åsberg Depression Rating Scale (MADRS), Clinical Global Impression Severity and Improvement scales (CGI-S and CGI-I); and C-SSRS. Additionally, subjects completed self-rated forms for depression (SDQ), anxiety (ASQ), cognition (Neuro-QoL), positive and negative affect (PANAS); potential confounders questionnaire (PCQ); quality of life (Q-LES-Q-SF); adverse events (SAFTEE); perceived blinding questionnaire (PBQ);

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and t-PBM self-report questionnaire (TSRQ). A study clinician monitored subjects for depressive worsening, adverse events (AEs), or changes to concomitant medications.

t-PBM session: All study subjects underwent 4 t-PBM sessions in 3T Siemens Trio MRI scanners at NKI and MGH, 1 week apart, in random order, with: 1) Sham; 2) High dose: Pulse wave (PW), peak irradiance: ~900 mW/cm^2, average irradiance ~300 mW/cm^2, 40Hz, 600 s, 33% duty cycle, 4.3 kJ total energy; 3) Medium dose: Continuous Wave (CW), ~300 mW/cm^2 irradiance, 333 s, 2.4 kJ total energy; 4) Low dose: CW, ~50 mW/cm^2 irradiance; 1200 s, 1.4 kJ total energy. t-PBM doses are differentiated by total energy and peak irradiance. Other t-PBM parameters were kept unchanged for all doses: 808 nm; 12.0 cm2 × 2 treatment area on each side (total 24 cm^2); delivered to the prefrontal cortex, bilaterally; the anatomical sites targeted include the dlPFC (standard EEG electrode sites F4, F3), mPFC (Fz, Fpz) and frontal pole (Fp2, Fp1). The study device was t-PBM-2.0, an investigational device (manufactured by LiteCure LLC, New Castle, DE) based on LiteCure's LightForce® EXPi Deep Tissue Laser Therapy™ System. For the investigational study, the EXPi System's beam delivery was fitted with an MRI-compatible helmet and modified to remain in stand-by, with a visible light but not delivering NIR, for the sham mode.

Imaging was performed on 3T Siemens Trio MRI scanners utilizing 12-channel transmit/receive head coils (the t-PBM device fit under the coil). After localization, we acquired structural images with a T1-weighted MPRAGE sequence (FOV 256 mm, in plane resolution 256 x 256, 192 1 mm slices, TR = 2500 ms, TE = 3.5 ms, TI = 1200 ms, flip angle 8°). Multi-echo resting state fMRI was recorded pre-, during, and immediately post t-PBM/sham. The parameters were: 2174 Hz/Px bandwidth, TR = 2500 ms, TE1 = 12.8 ms, TE2 = 33.41 ms, TE3 = 54.02 ms, voxel size 2.5 mm × 2.5 mm x 2.5 mm, 60 slices, multiband acceleration factor = 2, flip angle 65°.

Preprocessing of MRI and fMRI data was performed with the AFNI software package [20]. Preprocessing steps included: despiking (removing gross spikes from the time series of each voxel), slice time correction to bring each 2D image onto the same timestamp, alignment of the functional volumes onto the anatomical volume, spatial smoothing with a FWHM of 4 mm, masking out all non-brain voxels. Each voxel's time series was scaled by a single multiplicative constant so that the full-scan series had a mean of 100, allowing subsequent values to be interpreted as percent signal change; because this constant is identical across the pre-, during-, and post-stimulation segments of each voxel, it leaves the ratio of BOLD power between segments unchanged. We then regressed out nuisance signals, including the six rigid-body head-motion parameters and any signal outside the 0.01-0.1 Hz passband. Motion and other artifacts were censored at two levels: whole volumes for which AFNI flagged >50% of voxels as artifactual were removed, as were individual samples deviating > 4 SD from their voxel mean (~5% of data censored overall). The three echoes were preprocessed and analyzed separately rather than combined; no multi-echo ICA denoising was applied, and echo time was instead included as a fixed factor in all statistical models. Preprocessing was performed once on each full continuous run; the pre-, during-, and post-stimulation segments were extracted afterward, so that identical preprocessing was applied to all three periods and to all 4 doses.

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BOLD power was defined as the summed spectral power of the mean ROI time series within the 0.01–0.1 Hz low-frequency band; specifically, the sum of the power spectrum across this low frequency was calculated and taken to represent the total power of the BOLD signal.

*Image Processing:*BOLD signal change was measured for each pair of cortical ROIs defined by the Destrieux atlas [21] (75 regions in each hemisphere). We separated the resulting time series into pre-, during, and post-stimulation segments. We then performed spectral analysis to measure the BOLD power during each segment, employing the Thomson multitaper technique [22] to increase the signal-to-noise ratio of the ensuing power estimates. This produced three spectra for each echo and dose (averaged for before, during, and after t-PBM stimulation).

Statistical analyses: Demographic information was reported as counts (N), percentages, and means (standard deviations). The primary, pre-specified analysis focused on dose-dependent changes in BOLD power within the bilateral middle frontal gyrus (MFG; two brain regions). Additional exploratory analyses were conducted for the irradiated prefrontal cortex (14 regions) and the whole cortex (150 regions).

For each of the 3 analyses (MFG, frontal and entire cortex), a linear mixed-effects model was fitted using maximum likelihood (REML), with BOLD power as the response variable to analyze the effects of t-PBM irradiance doses on BOLD power. Initially, intercepts, slopes and variances from random (repeated) variables were explored. Subsequently, the final model was selected after comparing linear mixed-effects models fit using maximum likelihood (ML), with model selection based on predictions using the Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and deviance statistics. The final model included t-PBM treatment (sham, low, medium, and high doses), time (before and during the treatments), echo time (three distinct values for imaging), the brain regions, and their interactions as the fixed factors. Covariates included age, gender, MADRS scores, current antidepressant use (yes/no), and MDD recurrent or single episode. The repeated measurements in participants, and t-PBM dose sequence (“treatment week”) were included as crossed random effects to account for carry over effects. Whole cortex BOLD calculations only included all cortical brain regions (n = 150) as crossed random effect.

For each of the 3 analyses (MFG, frontal, and whole cortex), overall inference was based on a single omnibus test of the fitted mixed-effects model. When significant interaction effects were identified, post hoc pairwise comparisons were performed using estimated marginal means with Tukey-adjusted p values to account for multiple testing. Model assumptions were evaluated through residual diagnostics, and marginal and conditional R^2 values [23] were calculated to quantify explained variance. BOLD power values were natural log-transformed before analysis to improve distributional properties. Effect sizes were calculated relative to the pre-treatment variance of the sham condition when statistically significant effects were observed. All statistical analyses were conducted using the “lme4”, “lmerTest”, “emmeans”, “ggplot2”, and “easystats” packages in R software [24,25]. Statistical significance was defined as $p \leq 0.05$.

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3. Results

The study sample included 31 participants diagnosed with MDD, including 20 females and 11 males, most of whom underwent 4 fMRI scans. The mean age was 37.0 years (SD = 15.4). 28 participants had recurrent MDD, 3 had single episode MDD. Twenty-three participants were undergoing antidepressant treatment, while 8 were untreated. The mean baseline MADRS score was 27.6 (SD = 5.90), which decreased to 18.0 (SD = 9.46) at endpoint. The demographic data for all 31 subjects are presented in Table 1.

114 fMRI scans yielded valid data. Of the 10 scans not analyzable, 4 had partially missing data and 6 were rejected due to the pre-established artefact criteria (5% data loss due to artifacts). Twenty-four participants had all four fMRI scans with analyzable data. Seven participants had partially missing data on some of the four fMRI scans and provided 18 analyzable fMRI scans (of 28 scans recorded). No statistically significant difference was observed among the BOLD power obtained across study centers (MGH and NKI/NYU) ($F(_{2, 27.94}) = 1.376$, $p = 0.269$). Since most of the post-stimulation BOLD comparisons were not significantly different from sham (see Supplemental Material section 1 for details), we focused on pre-stimulation and during-stimulation periods for all comparisons below.

3.1. Effect of t-PBM on fMRI BOLD power in the middle frontal gyrus, bilateral

Our pre-specified analysis focused on the dose-dependent effects of t-PBM on the BOLD power in the middle frontal gyrus (MFG), bilaterally, based on simulations indicating MFG was in the center of the irradiated areas. The analyses included two cortical areas (left and right MFG) in the Destrieux atlas [21].

The model fixed effects explained 35.5% of the variance in the BOLD power within MFG, while the full model explained 69%, and there was a significant t-PBM Doses × Time interaction ($F(3, 1265.99) = 37.81$, $p < 0.0001$), implying that the treatment effect on BOLD power differed across time points based on t-PBM doses (see Supplemental Material sections 2A and 2B for the main and interaction effects). The details of estimated means of MFG BOLD power changes and their effect sizes for the four t-PBM doses are presented in Table 2. The low t-PBM dose was significantly associated with a decrease in MFG BOLD power across all echo times, while the medium t-PBM dose was significantly associated with an increase in MFG BOLD power at echo times 1 and 2 (compared to pre-treatment levels). Fig. 1 depicts the MFG ΔBOLD power (during-t-PBM minus pre-t-PBM BOLD power) for each t-PBM dose and participant. The low t-PBM dose was associated with the largest decrease in MFG BOLD power at echo time 1 (Cohen's $d = -0.69$, Table 2). The medium t-PBM dose was associated with an increase in MFG BOLD power at echo time 1 (Cohen's $d = 0.56$, Table 2). The sham condition and the high t-PBM dose did not show a significant mean difference between during- and pre-treatment across all echo times.

3.2. Effect of t-PBM on fMRI BOLD power in the prefrontal cortex directly illuminated by the laser

A simulation of the area of the prefrontal cortex directly illuminated by the laser included the following structures: middle frontal gyrus, middle frontal sulcus, superior frontal gyrus,

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superior frontal sulcus, inferior frontal sulcus, triangular part of the inferior frontal gyrus, transverse frontopolar gyri and sulci. Thus, the following analysis included 14 cortical areas from the Destrieux atlas (7 on each side).

The fixed effects and the full model R^2 explained 35.3% and 65.8% of variance in the BOLD signal within the prefrontal cortex, respectively (see Supplemental Material sections 3A and 3B for the main and interaction effects). Linear mixed-effects analysis showed a significant t-PBM Doses × Time interaction ($F_{(3, 1265.99)} = 37.81$, $p < 0.0001$). The estimated mean BOLD power changes in the prefrontal cortex with the four t-PBM doses are shown in Table 3. The medium t-PBM dose was significantly associated with an increase in prefrontal cortical BOLD, whereas low dose was significantly associated with a decrease in prefrontal cortical BOLD, compared to pre-treatment. Fig. 2 summarizes prefrontal cortical Δ (Δ = during minus pre-t-PBM) BOLD power associated with t-PBM doses in each participant. The low t-PBM dose was associated with the largest decrease in prefrontal BOLD power at echo time 1 (Cohen's d = −0.70, Table 3). The medium t-PBM dose was associated with a large increase in prefrontal BOLD (d = 0.52, Table 3). The BOLD changes from the high dose t-PBM and sham were not significantly different from zero in all echo times. Fig. 3 summarizes the observed BOLD differences between during- and pre t-PBM intervention in prefrontal cortex.

3.3. Effect of t-PBM on BOLD power in the entire cortex

A similar analysis was performed for the effect of t-PBM on BOLD power change in the entire brain cortex (the majority was not directly illuminated by t-PBM). The analysis included all 150 cortical areas from the Destrieux atlas (75 on each side). See Supplemental Material sections 4A and 4B for the main and interaction effects.

The linear mixed-effect model showed a significant interaction effect of t-PBM by time on prefrontal cortical BOLD power ($F_{(3, 100599.9)} = 2236.37$, $p < 0.0001$). The details of estimated means and changes in global cortical BOLD power across the four t-PBM doses are shown in Table 4. The medium t-PBM dose was significantly associated with an increase in global cortical BOLD power, while the low dose was significantly associated with a decrease in global cortical BOLD power, compared to pre-treatment, at all echo times. The high dose t-PBM was associated with a small decrease in global cortical BOLD power, only at echo times 2 and 3. Fig. 4 summarizes global cortical Δ BOLD power (during-minus pre-treatment BOLD power) induced by t-PBM doses in each participant. The low t-PBM dose was linked to the largest decrease in global cortical BOLD power at echo time 1 (Cohen's d = −0.62, Table 4). The medium t-PBM dose was associated with a substantial increase in global cortical BOLD at echo 1 (Cohen's d = 0.48, Table 4). The high t-PBM dose was associated with a small decrease in global cortical BOLD power (largest at echo time 3: Cohen's d = 0.12, Table 4). The BOLD power mean change associated with sham was not significantly different from zero across all echo times.

3.4. Clinical effects (efficacy, tolerability and side effects)

As previously reported [26], participants remained blinded during treatments 1-2 of the series, but blinding decreased during treatments 3-4 to all active t-PBM doses, with no

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significant differences in blinding between t-PBM doses. See Supplemental Material section 7 for details.

We detected no significant dose-specific antidepressant effects after single t-PBM administration. Changes in the severity of depression (MADRS) with each t-PBM dose were calculated from the rating prior to each t-PBM administration to one week later. Results from fitting a linear mixed-effect model indicated that there was no main effect of the single administration of t-PBM doses on MADRS. The fixed effects model explained 18.5% of the variance in MADRS, while the full model explained 71.9%. There was no statistically significant interaction effect of t-PBM by time on MADRS scores ($F_{(3,\ 195.72)} = 1.858$, $p = 0.1379$). See Supplemental Material section 5 for details.

t-PBM was very well tolerated. There were no SAEs and no AEs rated as both severe and likely or possibly associated with the study treatment. Based on the SAFTEE, there were no statistically significant changes in rates of any side effects, for any of the t-PBM doses and sham, compared to baseline or the prior week. A detailed report of side effects analyses was published [26]. Also, using proton magnetic resonance spectroscopy (1H-MRS) to measure brain temperature, we found no increases in brain temperature with any of the t-PBM doses [27].

4. Discussion

Our findings strongly suggest that t-PBM has a robust, dose-dependent effect on cortical BOLD power, and can significantly modulate brain activity.

During t-PBM stimulation we found t-PBM dose- and time-dependent changes in BOLD power in the MFG, prefrontal cortex, and the entire brain cortex. In the MFG, both low and medium t-PBM doses significantly altered the BOLD signal, but in opposing directions. Both the medium dose (continuous wave, 300 mW/cm^2) and the low t-PBM dose (continuous wave, 50 mW/cm^2) were associated with medium to large effect size changes in BOLD power, albeit in opposite directions. The effects of the medium t-PBM dose (Cohen's $d = 0.56$) in MFG were consistent with our initial hypothesis and represent a demonstration of target engagement. The low t-PBM dose was associated with a decrease in MFG BOLD across all echo times (Cohen's $d = -0.69$), while the medium dose resulted in a significant increase in BOLD at echo times 1 and 2. However, at echo time 3, the effect of the medium dose was marginally significant (the study had insufficient power to detect a fully significant effect at this Echo time). In the prefrontal cortical area irradiated, the effect sizes of BOLD power changes were similar to those observed in the MFG, with an increase in BOLD with the medium t-PBM dose (Cohen's $d = 0.52$) and decrease in BOLD with the low t-PBM dose ($d = -0.70$), across all echo times.

In contrast to our prediction, the low t-PBM dose was associated with a decrease in BOLD. Since the BOLD signal is both the result of complex interaction between cerebral blood flow and metabolism, a BOLD decrease is consistent with increased brain metabolism (i.e., increased consumption of oxygen and production of deoxygenated hemoglobin) without a matched increase in blood flow. While this mechanistic hypothesis will need further

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investigation, it appears the low t-PBM dose is indeed biologically active, although with different mechanisms from the middle dose.

We observed no significant effects on BOLD with the high (pulsed) t-PBM dose in either the MFG or the prefrontal cortical areas illuminated. The very high peak irradiance (900 mW/cm^2) in our pulsed dose may have a negative impact on cortical excitability [28], in contrast to lower irradiance, which could explain its lack of impact on the BOLD signal. It is also possible the difference may be primarily related to the waveform (pulsed, versus continuous in the middle dose). Of note, we initially considered a continuous wave 700 mW/cm^2 as the high dose, but it was very poorly tolerated, with most (9 of 10) participants receiving a test dose reporting difficult to tolerate skin warming and requesting to abort the test. We therefore selected the pulsed dose, which was well tolerated and can provide the same total energy per session, for the fMRI experiment.

We confirmed our pre-specified hypothesis by detecting changes in BOLD power in the MFG (located near the center of the illuminated area). We also detected changes in BOLD, similar in magnitude and direction, in the larger prefrontal area illuminated, and in the entire brain cortex (the majority of which is not directly illuminated by the laser). This unexpected finding is not likely explained by non-specific factors (scanner drift was removed by the 0.01-0.1 Hz band-pass, head motion was regressed and censored, and cardiac and respiratory rhythms fall outside the analysis band), but this cannot be definitely ruled out. The whole-cortex effect was dose-specific and bidirectional, a pattern inconsistent with non-specific confounds. If replicated, the effects of t-PBM on the entire brain cortex - possibly related to the release of vasodilating nitric oxide - would be very important, allowing targeting brain pathology outside the prefrontal areas even while optimal transmission through the skull (and avoiding the diffractive role of hair) supports t-PBM application to prefrontal areas.

We found no significant dose-dependent effects of single t-PBM administrations on depression severity (MADRS). This was anticipated, as the antidepressant effects of t-PBM have been reported after repeated administrations over 6–12 weeks. This aligns with the notion that neuroplastic changes, which are critical for long-term mood improvements, may require more time (and repeated t-PBM administrations) to develop, compared to the acute changes in brain activity observed in this study. t-PBM was well tolerated by participants, with no significant adverse events reported at the doses tested. This supports the safety of t-PBM as a non-invasive treatment for MDD subjects.

Our study has several limitations. It was primarily designed to detect immediate biological effects of t-PBM and it does not provide information for longer-term effects. As a single dose of t-PBM is likely not sufficient to generate clinical improvement, we were unable to evaluate correlations between biological and clinical effects. Future studies with repeated administration will be required. Our conclusions are limited to the three t-PBM doses tested, which differ in waveforms, peak irradiance, total energy, and stimulation duration. Given the divergent effects we detected at different doses, it is not clear how to extrapolate these results to other irradiance doses. Additional studies, testing a larger number of parameter groups will be required. Our determination of the area illuminated by the laser is based on modeling and may not reflect exactly the individualized boundary of the optical exposure

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field or the downstream biological effects. The sham condition was not matched to active stimulation, and blinding decreased during treatments 3-4. The study cross-over design was another potential limitation, as it assumes there are no carryover effects of a single administration of t-PBM at 1 week or beyond. Fortunately, this assumption is consistent with prior literature and with our data, as we did not detect an order effect for t-PBM doses. We did not acquire simultaneous cardiac or respiratory recordings to apply model-based physiological noise correction, although such signals lie largely outside our 0.01-0.1 Hz analysis band.

5. Conclusion

In conclusion, we observed a robust and consistent increase in cerebral BOLD power with the medium t-PBM dose and a significant and consistent decrease in cerebral BOLD power with the low t-PBM dose, particularly in regions involved in emotional regulation and executive functioning, such as the middle frontal gyrus (MFG) and the prefrontal cortex. Surprisingly, similar effects on BOLD were also detected in the entire cortex, beyond the areas directly illuminated by the t-PBM laser. These results are important for dose optimization in future clinical t-PBM studies.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Declaration of competing interests

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Dan V Iosifescu reports financial support was provided by National Institute of Mental Health. Dan V Iosifescu reports a relationship with Autobahn Therapeutics, Inc. that includes: consulting or advisory. Dan V Iosifescu reports a relationship with Axsome Therapeutics Inc that includes: consulting or advisory. Dan V Iosifescu reports a relationship with Boehringer Ingelheim Pharmaceuticals Inc that includes: consulting or advisory. Dan V Iosifescu reports a relationship with Clexio Biosciences Ltd that includes: consulting or advisory. Dan V Iosifescu reports a relationship with Delix that includes: consulting or advisory. Dan V Iosifescu reports a relationship with MindMed that includes: consulting or advisory. Dan V Iosifescu reports a relationship with Neumora Therapeutics Inc that includes: consulting or advisory. Dan V Iosifescu reports a relationship with Relmada Therapeutics Inc that includes: consulting or advisory. Dan V Iosifescu reports a relationship with SAGE Therapeutics Inc that includes: consulting or advisory. Luis De Taboada reports a relationship with Monsabi Inc that includes: employment. Paolo Cassano reports a relationship with Janssen Pharmaceuticals Inc that includes: consulting or advisory. Paolo Cassano reports a relationship with Niraxx Inc that includes: consulting or advisory and equity or stocks. Paolo Cassano has patent issued to Massachusetts General Hospital and Niraxx Inc. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.brs.2026.103185.

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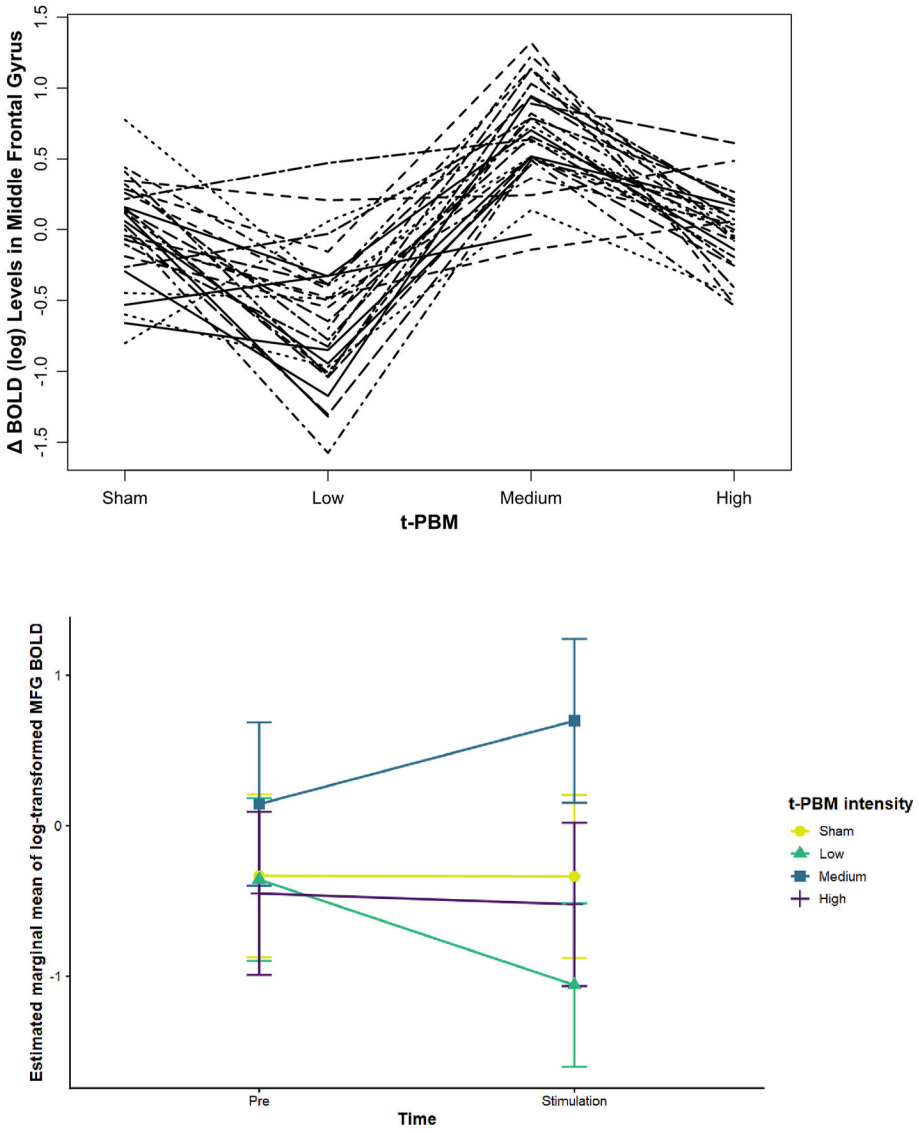


Fig. 1.

a) Middle frontal gyrus BOLD power changes in each participant by t-PBM doses. Each line represents the participant's Δ values (during minus pre-t-PBM BOLD power) at each t-PBM dose. **b)** Estimated means of log-transformed MFG BOLD values by time and t-PBM intensity. Estimated means were obtained from the linear mixed-effects regression model adjusted for age, MADRS score, gender, single versus recurrent major depressive episode, current antidepressant use, brain region (left or right), and echo time, with random intercepts for participant and treatment week. Points represent model-estimated marginal means, lines connect the estimated marginal means across time within each t-PBM intensity group, and error bars indicate 95% confidence intervals for the estimated marginal means.

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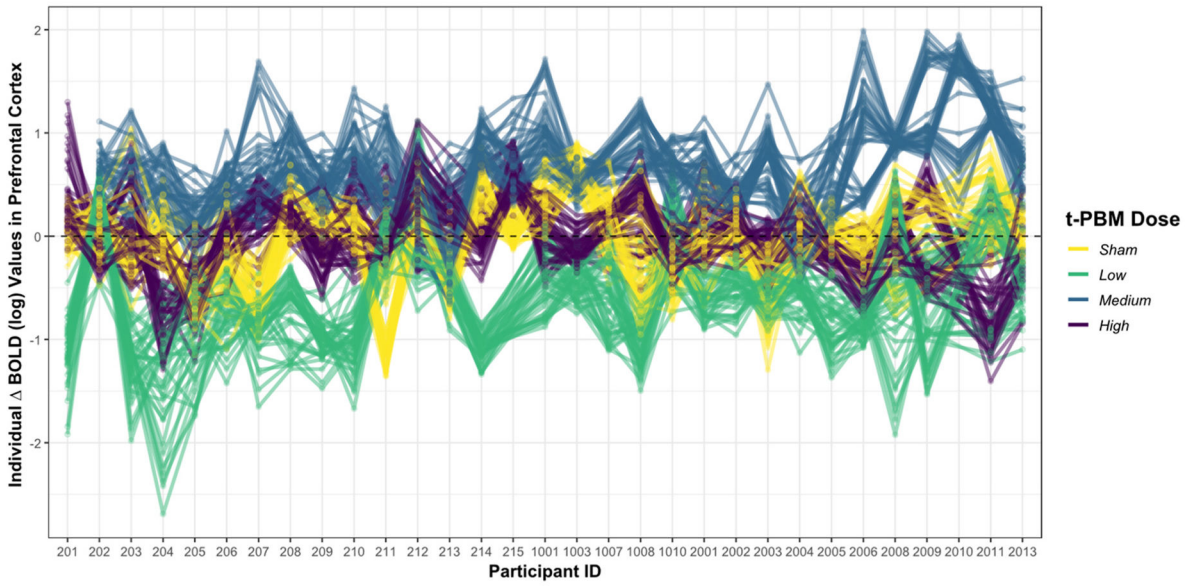


Fig. 2.

Δ BOLD power (during minus pre-t-PBM) in prefrontal areas by t-PBM dose for all participants (connected dots represent prefrontal brain regions (7), sides (2) and echo times (3) for each participant).

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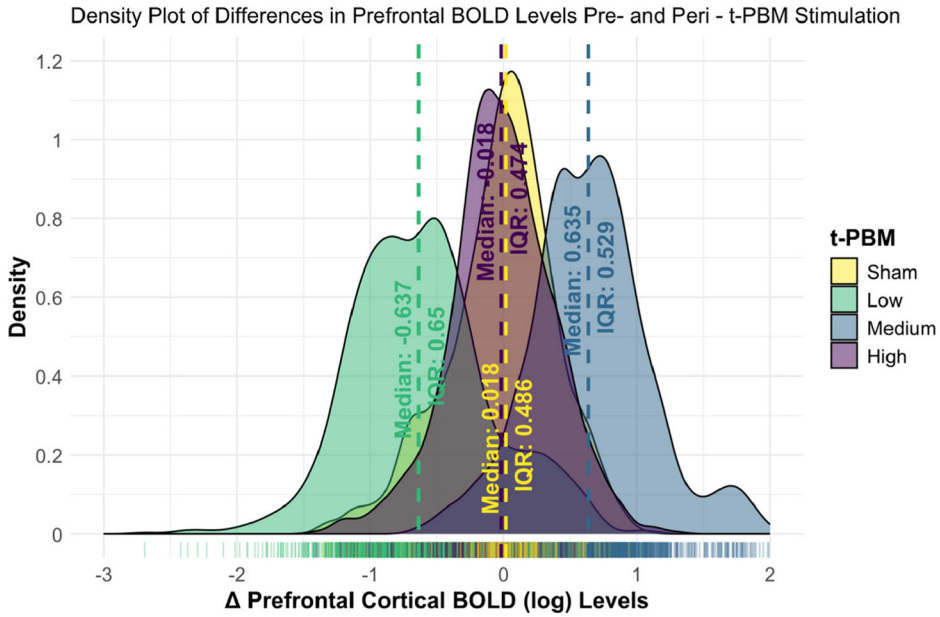


Fig. 3.

Density plots of Δ BOLD power (during minus pre-t-PBM) in prefrontal areas, by t-PBM doses.

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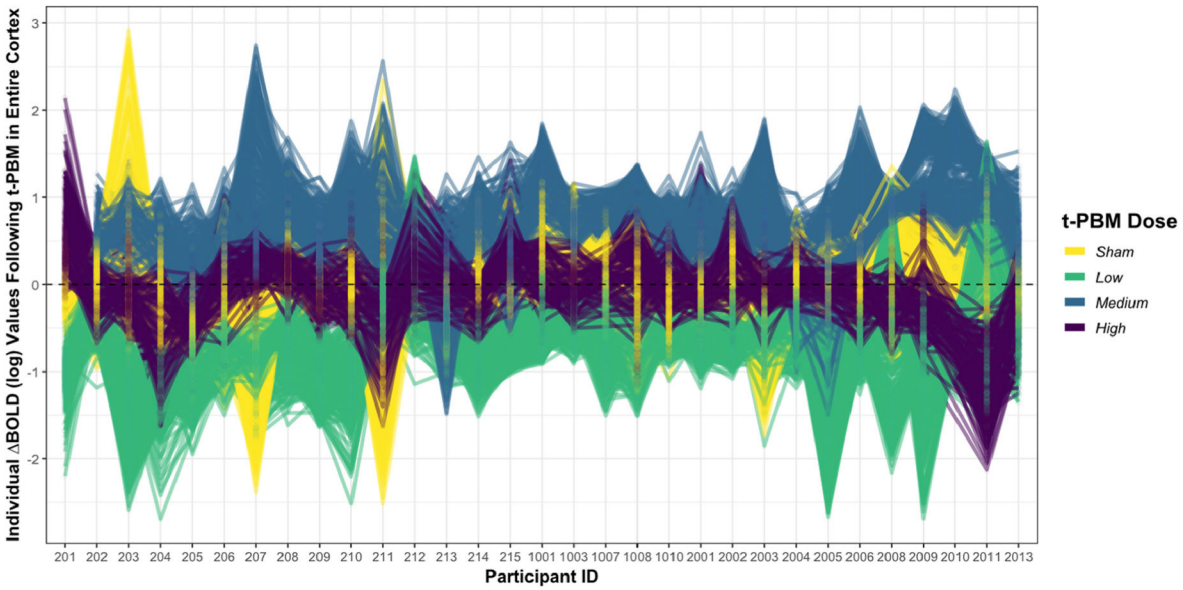


Fig. 4.

Global cortical BOLD power changes in each participant by t-PBM doses. Each line represents a participant's Δ BOLD power for each t-PBM dose (Connected dots represent brain regions (75), sides (2) and echo times (3) for each participant. Horizontal lines represent Loess regression for each t-PBM dose).

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Table 1

Demographic data of study participants.

	Treatment Group			
	t-PBM at High Irradiance	**t-PBM at Medium Irradiance**	**t-PBM at Low Irradiance**	**Sham**
Participants with scans	27	29	29	29
Age Mean (SD)	37.37 (15.61)	36.45 (15.65)	37.24 (15.67)	37.24 (15.67)
Sex: Female	17	19	20	19
Male	10	10	9	10
Race: White	18	18	19	18
Black or African American	3	3	3	3
Asian	3	4	3	4
Hispanic/Latino	1	1	2	2
Multiple	1	2	1	1
Other	1	1	1	1
Change in MADRS Total Score (Baseline - After) Mean (SD)	4.30 (8.44)	6.76 (8.98)	6.21 (8.20)	2.97 (7.76)
Change in ASQ Total Score (Baseline-After) Mean (SD)	33.67 (51.38)	42.6 (56.44)	38.69 (63.81)	26.97 (48.95)
Baseline Data for all 31 subjects				
Baseline BMI (kg/m^2) - Mean (SD)	26.1 (7.18)			
Length of MDD episode (months) - Mean (SD)	16.53 (21.49)			
Baseline MADRS score - mean (SD)	27.6 (5.90)			
Baseline IDS-C score - mean (SD)	32.94 (6.44)			
Baseline ASQ score - mean (SD)	132.03 (59.55)			
Baseline C-SSRS total score - mean (SD)	0.7 (0.83)			

BMI = body mass index; CGI-S= CGI-S=Clinical Global Impression - Severity; MADRS= Montgomery-Åsberg Depression Rating Scale; ASQ = Anxiety Symptom Questionnaire; IDS-C = Inventory of depressive symptomatology (clinician rating); C-SSRS= Columbia Suicide Severity Rating Scale (Suicidal Ideation); SD = standard deviation.

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Table 2

Estimated means of BOLD power change in middle frontal gyrus, MFG (during minus pre-t-PBM).

BOLD levels in MFG		Comparisons of Mean Changes in Treatment Arms[a]					Effect Size Calculations and 95% CI[b]				
Echo Times	**Contrast (During minus Pre BOLD)**	**Estimate**	**SE**	**df**	**t ratio**	**p value**	**Cohen's d**	**SE**	**df**	**lower.CL**	**upper.CL**
Echo	Sham	0.042014155	0.140089752	1262.119	0.29991	1.000000	0.03907	0.13029	35.6638	−0.225252	0.3034
Time	Low	−0.741327184	0.141377666	1264.911	−5.243595	0.000049	−0.689455	0.1322	35.43619	−0.957718	−0.421192
1	Medium	0.60597862	0.142231317	1264.896	4.26051	0.005024	0.56358	0.132729	36.28868	0.29446	0.83269
	High	−0.023186012	0.144262268	1262.992	−0.160721	1.000000	−0.021564	0.134169	36.20367	−0.293617	0.25049
Echo	Sham	−0.007197818	0.140089752	1262.119	−0.05138	1.000000	−0.006694	0.130287	35.6638	−0.271016	0.25763
Time	Low	−0.677634858	0.141377666	1264.911	−4.793083	0.000467	−0.630219	0.132083	35.43619	0.898244	0.362195
2	Medium	0.545607965	0.142231317	1264.896	3.83606	0.025956	0.50743	0.132665	36.28868	0.23845	0.77641
	High	−0.097821468	0.144262268	1262.992	−0.678081	1.000000	−0.090977	0.13418	36.20367	−0.363054	0.1811
Echo	Sham	−0.048724715	0.140089752	1262.119	−0.347811	1.000000	−0.045315	0.130291	35.6638	−0.309643	0.21901
Time	Low	−0.684189725	0.141377666	1264.911	−4.839447	0.000374	−0.636316	0.131578	35.43619	−0.903316	−0.369315
3	Medium	0.508015297	0.142231317	1264.896	3.57175	0.063475	0.47247	0.132613	36.28868	0.20359	0.74135
	High	−0.098119427	0.144262268	1262.992	−0.680146	1.000000	−0.091254	0.13418	36.20367	−0.363331	0.18082

[a]Results are averaged over the levels of gender, single or recurrent MDE, current antidepressant use, brain region. Degrees-of-freedom method: Kenward-Roger. P value adjustment: Tukey method for comparing a family of 24 estimates.

[b]Results are averaged over the levels of gender, single or recurrent MDE, current antidepressant use, brain region. Standard Deviation used for effect sizes: 1.075. Degrees-of-freedom method: inherited from Kenward-Roger when re-gridding. Confidence level used: 0.95.

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Table 3

Estimated means of BOLD power change in prefrontal cortex (during minus pre-t-PBM) by echo time.

BOLD levels in Prefrontal Cortex		Comparisons of Mean Changes in Treatment Arms[a]					Effect Size Calculations and 95%CI[b]				
Echo Times	**Contrast (during minus pre BOLD)**	**Estimate**	**SE**	**df**	**z ratio**	**p value**	**Cohen's d**	**SE**	**df**	**asymp.LCL**	**asymp. UCL**
Echo Time 1	Sham	0.01834	0.057441	Inf	0.31934	1.000000	0.0167	0.052301	Inf	−0.085807	0.11921
	Low	−0.768789	0.057974	Inf	−13.26092	0.000000	−0.699995	0.053042	Inf	−0.803956	−0.596034
	Medium	0.5761	0.05834	Inf	9.87502	0.000000	0.52455	0.053262	Inf	0.42016	0.62894
	High	−0.018482	0.059153	Inf	−0.31245	1.000000	−0.016829	0.05386	Inf	−0.122393	0.08874
Echo Time 2	Sham	−0.020586	0.057441	Inf	−0.358382	1.000000	−0.018744	0.052301	Inf	−0.121252	0.08376
	Low	−0.681256	0.057974	Inf	−11.75105	0.000000	−0.620295	0.052988	Inf	−0.724148	−0.516441
	Medium	0.4932	0.05834	Inf	8.45397	0.000000	0.44907	0.053224	Inf	0.34475	0.55338
	High	−0.085008	0.059153	Inf	−1.437083	0.999289	−0.077401	0.053863	Inf	−0.182971	0.02817
Echo Time 3	Sham	−0.044945	0.057441	Inf	−0.782458	1.000000	−0.040923	0.052302	Inf	−0.143434	0.06159
	Low	−0.675023	0.057974	Inf	−11.64354	0.000000	−0.61462	0.05288	Inf	−0.718262	−0.510977
	Medium	0.4837	0.05834	Inf	8.29109	0.000000	0.44042	0.05322	Inf	0.33611	0.54472
	High	−0.101768	0.059153	Inf	−1.720404	0.990772	−0.092661	0.053861	Inf	0.198226	−0.0129

[a]Results are averaged over the levels of gender, single or recurrent MDE, current antidepressant, brain region. Degrees-of-freedom method: asymptotic. P value adjustment: Tukey method for comparing a family of 24 estimates.

[b]Results are averaged over the levels of gender, single or recurrent MDE, current antidepressant use, brain region. Standard Deviation used for effect sizes: 1.098. Degrees-of-freedom method: inherited from asymptotic when re-gridding. Confidence level used: 0.95.

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Table 4

Estimated means of BOLD power change in the entire cortex (during minus pre-t-PBM) by echo time.

BOLD levels in Entire Cortex		Comparisons of Mean Changes in Treatment Arms[a]					Effect Size Calculations and 95%CI[b]				
Echo Times	**Contrast (During minus Pre BOLD)**	**Estimate**	**SE**	**df**	**z.ratio**	**p.value**	**Cohen's d**	**SE**	**df**	**asymp.LCL**	**asymp. UCL**
Echo Time 1	Sham	−0.017374	0.018476	Inf	−0.940366	1.000000	−0.013656	0.014523	Inf	−0.04212	0.01481
	Low	−0.792836	0.018649	Inf	−42.51412	0.000000	−0.623195	0.014724	Inf	−0.652054	−0.594336
	Medium	0.61162	0.018767	Inf	32.5912	0.000000	0.48076	0.014751	Inf	0.45184	0.50967
	High	−0.015043	0.019027	Inf	−0.790601	1.000000	−0.011824	0.014956	Inf	−0.041137	0.01749
Echo Time 2	Sham	−0.034884	0.018476	Inf	−1.888111	0.971548	−0.02742	0.014523	Inf	−0.055884	0.00104
	Low	−0.660411	0.018649	Inf	−35.41309	0.000000	−0.519104	0.014703	Inf	−0.547923	−0.490286
	Medium	0.48666	0.018767	Inf	25.9324	0.000000	0.38253	0.014776	Inf	0.35357	0.41149
	High	−0.086234	0.019027	Inf	−4.532185	0.001434	−0.067782	0.014957	Inf	−0.097097	−0.038468
Echo Time 3	Sham	−0.059362	0.018476	Inf	−3.212981	0.176502	−0.04666	0.014523	Inf	−0.075125	0.018196
	Low	−0.680293	0.018649	Inf	−36.47921	0.000000	−0.534732	0.014665	Inf	−0.563474	−0.50599
	Medium	0.51641	0.018767	Inf	27.5178	0.000000	0.40592	0.014779	Inf	0.37695	0.43488
	High	−0.123142	0.019027	Inf	−6.471961	0.000000	−0.096793	0.014957	Inf	−0.126109	−0.067477

Confidence level used: 0.95.

[a]Results are averaged over the levels of gender, single or recurrent MDE, current antidepressant use. Degrees-of-freedom method: asymptotic. P value adjustment: Tukey method for comparing a family of 24 estimates.

[b]Results are averaged over the levels of gender, single or recurrent MDE, currently antidepressant use. Standard deviation used for effect sizes: 1.272. Degrees-of-freedom method: inherited from asymptotic when re-gridding.